ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2025 P1H Worked Solutions

Jun 2025 | P1H

31 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 5

TOPIC: **Statistics Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	**Frequency**
$0 < d \leqslant 5$	26
$5 < d \leqslant 10$	40
$10 < d \leqslant 15$	16
$15 < d \leqslant 20$	10
$20 < d \leqslant 25$	8

(a) Write down the modal class.

....
(1)

(b) Work out an estimate for the mean distance.

.... km
(4)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 5

TOPIC: **Statistics Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Statistics Toolkit. The tag is correct.

Solution

The modal class is the class with the greatest frequency.
Modal class: $5 < d \leq 10$.
Use midpoints $2.5, 7.5, 12.5, 17.5, 22.5$.

$$2.5(26) + 7.5(40) + 12.5(16) + 17.5(10) + 22.5(8) = 920$$

$$\text{estimated mean} = \frac{920}{100} = 9.2$$

FINAL ANSWER

modal class $5 < d \leq 10$, estimated mean 9.2 km.

PAST PAPER QUESTION

Topic: **Percentages**

Jun 2025 P1H - Jun 2025 | P1H

Q#: 2 Marks: 6

2 Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

(a) Work out the percentage profit Anna makes.
Show your working clearly.

....%

(4)

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

....cm

(1)

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

....g

(1)

(Total for Question 2 is 6 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 6

TOPIC: **Percentages**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Percentages. Most of the marks are for percentage profit; bounds are secondary.

Method

Number of cups:

$$80 \times 4 = 320$$

Cost to make the cups:

$$320 \times 6 = 1920$$

Profit:

$$2160 - 1920 = 240$$

Percentage profit:

$$\frac{240}{1920} \times 100 = 12.5$$

The height 9 cm correct to the nearest cm has lower bound

$$8.5 \text{ cm}$$

The weight 120 g correct to the nearest 10 g has upper bound

$$125 \text{ g}$$

FINAL ANSWER

(a) 12.5%, (b) 8.5 cm, (c) 125 g

PAST PAPER QUESTION

Q#: 3 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2025 P1H - Jun 2025 | P1H

3

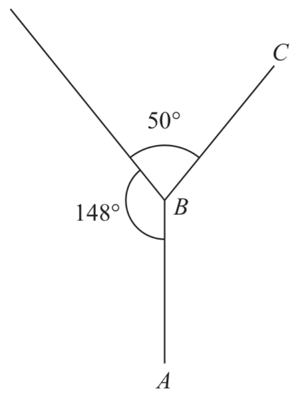

Diagram **NOT** accurately drawn

AB and *BC* are two sides of a regular polygon with *n* sides.

Work out the value of *n*
Show your working clearly.

n = ...

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

Angles around point B add to $360°$.
The interior angle of the regular polygon is

$$360° - 140° - 50° = 170°$$

So the exterior angle is

$$180° - 170° = 10°$$

For a regular polygon,

$$n = \frac{360}{10} = 36$$

FINAL ANSWER

$n = 36$.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P1H - Jun 2025 | P1H

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m =$ ....

(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$n =$ ....

(1)

(c) Simplify fully $(5a^4r^2)^3$

....

(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Algebraic Roots and Indices. The tag is correct.

Method

For part (a),

$$x^5 \times x^7 = x^{12}$$

So $m = 12$.
For part (b),

$$y^8 \div y^3 = y^{8-3} = y^5$$

So $n = 5$.
For part (c),

$$(5a^4r^2)^3 = 5^3a^{4\times3}r^{2\times3}$$

$$= 125a^{12}r^6$$

FINAL ANSWER

(a) $m = 12$, (b) $n = 5$, (c) $125a^{12}r^6$.

PAST PAPER QUESTION

Q#: 5 Marks: 3

TOPIC: **Percentages**

Jun 2025 P1H - Jun 2025 | P1H

5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.

Work out the normal price of the watch.

.... euros

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Percentages**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Corrected to Percentages. This is a reverse percentage sale-price question.

Method

The watch is reduced by 28%, so the sale price is

$$100\% - 28\% = 72\%$$

of the normal price.

$$0.72 \times \text{normal price} = 198$$

$$\text{normal price} = 198 \div 0.72 = 275$$

FINAL ANSWER

275 euros.

PAST PAPER QUESTION Q#: 6 Marks: 6

TOPIC: **Factorising**

Jun 2025 P1H - Jun 2025 | P1H

6 (a) Solve $x - 4 = \dfrac{3 + 2x}{6}$

Show clear algebraic working.

$x =$ ....

(3)

(b) (i) Factorise $y^2 - 11y + 30$

....

(2)

(ii) Hence solve $y^2 - 11y + 30 = 0$

....

(1)

(Total for Question 6 is 6 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 6

TOPIC: **Factorising**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is acceptable because the factorising and solving-quadratic section is a full half of the question.

Solution

(a)

$$x - 4 = \frac{3 + 2x}{6}$$

$$6x - 24 = 3 + 2x$$

$$4x = 27$$

$$x = \frac{27}{4}$$

(b)(i)

$$y^2 - 11y + 30 = (y - 5)(y - 6)$$

(b)(ii)

$$y = 5 \quad \text{or} \quad y = 6$$

FINAL ANSWER

$x = \frac{27}{4}$, $(y - 5)(y - 6)$, $y = 5$ or $y = 6$

PAST PAPER QUESTION Q#: 7 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P1H - Jun 2025 | P1H

7 Here is a solid cylinder.

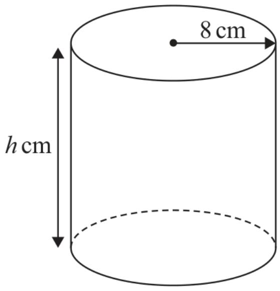

Diagram **NOT** accurately drawn

The radius of the cylinder is 8 cm
The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h
Give your answer correct to one decimal place.

h = ...

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Volume & Surface Area. The tag is correct.

Solution

For a cylinder,

$$V = \pi r^2 h$$

$$3892 = \pi (8)^2 h$$

$$h = \frac{3892}{64\pi} = 19.357\ldots$$

FINAL ANSWER

19.4 cm.

PAST PAPER QUESTION Q#: 8 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1H - Jun 2025 | P1H

8 (a) Write 520 million in standard form.

....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

....
(1)

(c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

....
(2)

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$520 \text{ million} = 520000000 = 5.2 \times 10^8$$

$$8.79 \times 10^{-5} = 0.0000879$$

For part (c),

$$(5 \times 10^{42})(7 \times 10^{-180}) = 35 \times 10^{-138}$$

Write this in standard form:

$$35 \times 10^{-138} = 3.5 \times 10^{-137}$$

FINAL ANSWER

(a) 5.2×10^8, (b) 0.0000879, (c) 3.5×10^{-137}

PAST PAPER QUESTION Q#: 9 Marks: 5

TOPIC: **Area & Perimeter**

Jun 2025 P1H - Jun 2025 | P1H

9

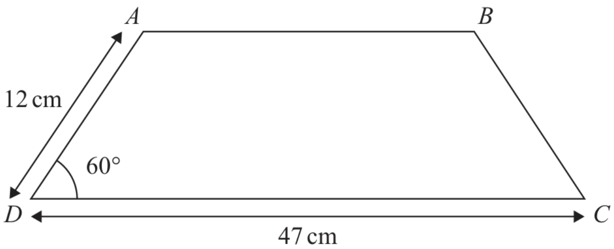

Diagram **NOT** accurately drawn

ABCD is a trapezium with one line of symmetry.

angle $ADC = 60°$ $AD = 12$ cm $DC = 47$ cm

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^2

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 5

TOPIC: **Area & Perimeter**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

Because the trapezium has one line of symmetry, each sloping side makes the same 60° angle with the base.
Height:

$$12 \sin 60^\circ = 10.392\ldots$$

The top parallel side is

$$47 - 2(12 \cos 60^\circ) = 35$$

Area:

$$\frac{1}{2}(47 + 35)(10.392\ldots) = 426.084\ldots$$

FINAL ANSWER

426 cm^2.

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Graphs of Functions**

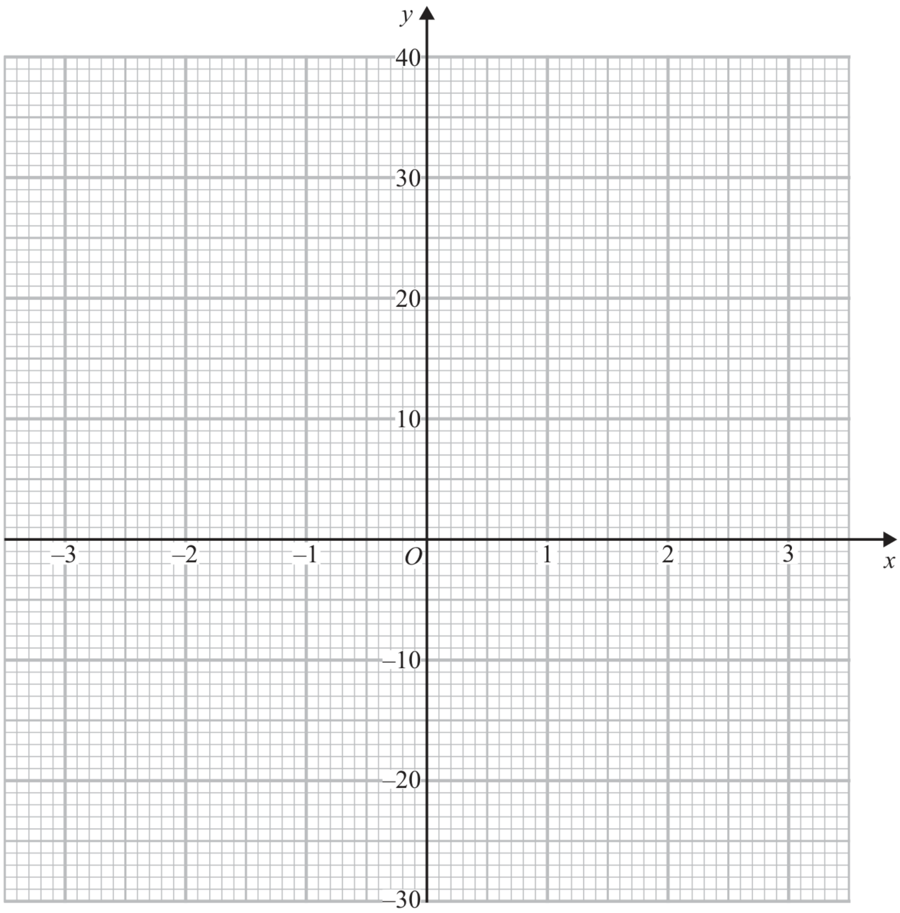
Jun 2025 P1H - Jun 2025 | P1H

10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	–3	–2	–1	0	1	2	3
y		–9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leqslant x \leqslant 3$



(2)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Graphs of Functions**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Correct. This is a table-and-graph question for a cubic function.

Solution

$$y = x^3 + 2x + 3$$

The completed table is:

x	-3	-2	-1	0	1	2	3
y	-30	-9	0	3	6	15	36

Plot the points and join them with a smooth cubic curve.

FINAL ANSWER
Missing values are $-30, 15$.

PAST PAPER QUESTION

Q#: 11 **Marks: 2**

Topic: **Circles, Arcs & Sectors**

Jun 2025 P1H - Jun 2025 | P1H

11 *OABC* is a sector of a circle, centre *O*

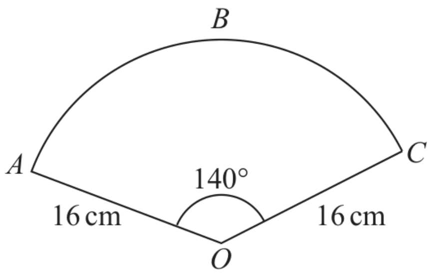


Diagram **NOT** accurately drawn

angle $AOC = 140°$
$OA = OC = 16$ cm

Calculate the area of the sector.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 11 is 2 marks)

PAST PAPER SOLUTION Q#: 11 Marks: 2

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circles, Arcs & Sectors. The tag is correct.

Solution

$$\text{area} = \frac{140}{360} \times \pi(16)^2$$

$$= 312.763\ldots$$

FINAL ANSWER

313 cm^2.

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2025 P1H - Jun 2025 | P1H

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

....

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected to Algebraic Fractions.

Solution

$$\frac{5}{4} + \frac{x-3}{6x}$$

The common denominator is $12x$:

$$\frac{5}{4} = \frac{15x}{12x}$$

$$\frac{x-3}{6x} = \frac{2x-6}{12x}$$

$$\frac{15x}{12x} + \frac{2x-6}{12x} = \frac{17x-6}{12x}$$

FINAL ANSWER

$\frac{17x-6}{12x}$.

PAST PAPER QUESTION

Q#: 13 **Marks: 3**

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P1H - Jun 2025 | P1H

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%
In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by x%

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

x = ...

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Compound Interest and Depreciation. The tag is correct.

Method

After the first two years, the value is

$$750000(0.96)(0.935) = 673200$$

Then the value increases by $x\%$ to 698445, so

$$673200\left(1 + \frac{x}{100}\right) = 698445$$

$$1 + \frac{x}{100} = \frac{698445}{673200} = 1.0375$$

$$x = 3.75$$

FINAL ANSWER

$x = 3.75$

PAST PAPER QUESTION

Q#: 14 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

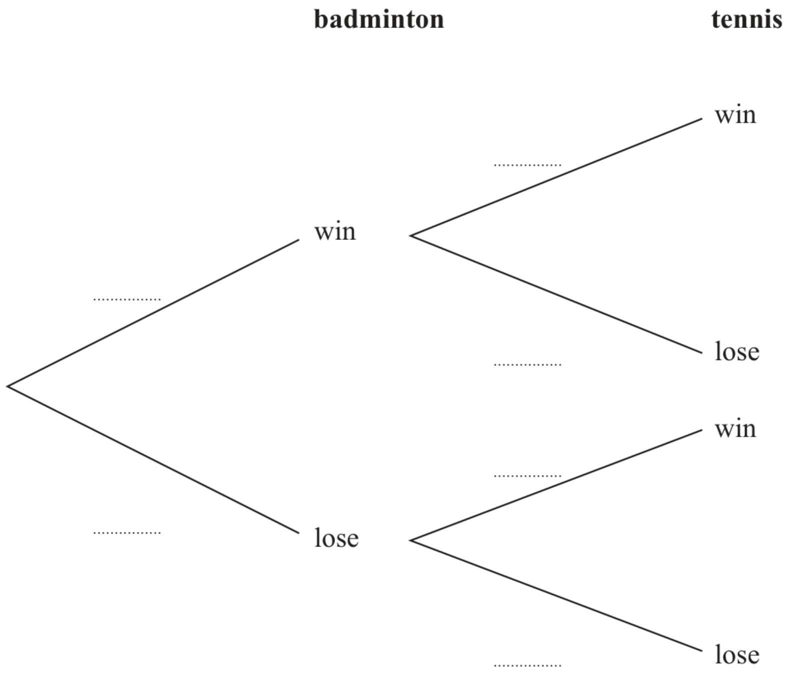
Jun 2025 P1H - Jun 2025 | P1H

14 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7
The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Ricardo loses both games.

....

(2)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

$$P(\text{loses badminton}) = 1 - 0.7 = 0.3$$

$$P(\text{loses tennis}) = 1 - 0.4 = 0.6$$

$$P(\text{loses both}) = 0.3 \times 0.6 = 0.18$$

FINAL ANSWER

0.18.

PAST PAPER QUESTION

Q#: 15 **Marks: 4**

TOPIC: **Algebraic Proof**

Jun 2025 P1H - Jun 2025 | P1H

15 $\left(\sqrt{3}\right)^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k =$ ....

(1)

(b) Show that $\dfrac{21}{3-\sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.
Show each stage of your working clearly.

(3)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 4

TOPIC: **Algebraic Proof**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic proof. The tag is acceptable because the question asks for exact-form proof.

Solution

(a)

$$(\sqrt{3})^5 = (\sqrt{3})^4\sqrt{3} = 9\sqrt{3}$$

So $k = 9$.

(b)

$$\frac{21}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{21(3+\sqrt{2})}{9-2}$$

$$= 3(3+\sqrt{2}) = 9 + 3\sqrt{2}$$

Since $3\sqrt{2} = \sqrt{18}$,

$$9 + 3\sqrt{2} = 9 + \sqrt{18}$$

FINAL ANSWER

$k = 9$, and $\dfrac{21}{3-\sqrt{2}} = 9 + \sqrt{18}$.

PAST PAPER QUESTION Q#: 16 Marks: 4

TOPIC: **Functions**

Jun 2025 P1H - Jun 2025 | P1H

16 $f(x) = 9 - \sqrt{x}$ where $x \geqslant 0$

$g(x) = 4x^2$

(a) Find f(9)

....

(1)

(b) Solve $fg(x) < 0$

Show clear algebraic working.

....

(3)

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Functions**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Corrected to Functions. The original tag was solving quadratic equations, but the question uses function notation and composition.

Solution

$$f(x) = 9 - \sqrt{x}, \qquad g(x) = 4x^2$$

(a)

$$f(9) = 9 - \sqrt{9} = 6$$

(b)

$$fg(x) = f(g(x)) = 9 - \sqrt{4x^2}$$

$$= 9 - 2|x|$$

Solve

$$9 - 2|x| < 0$$

$$2|x| > 9$$

$$|x| > \frac{9}{2}$$

So

$$x < -\frac{9}{2} \quad \text{or} \quad x > \frac{9}{2}$$

If the input is restricted to $x \geq 0$, this becomes $x > \frac{9}{2}$.

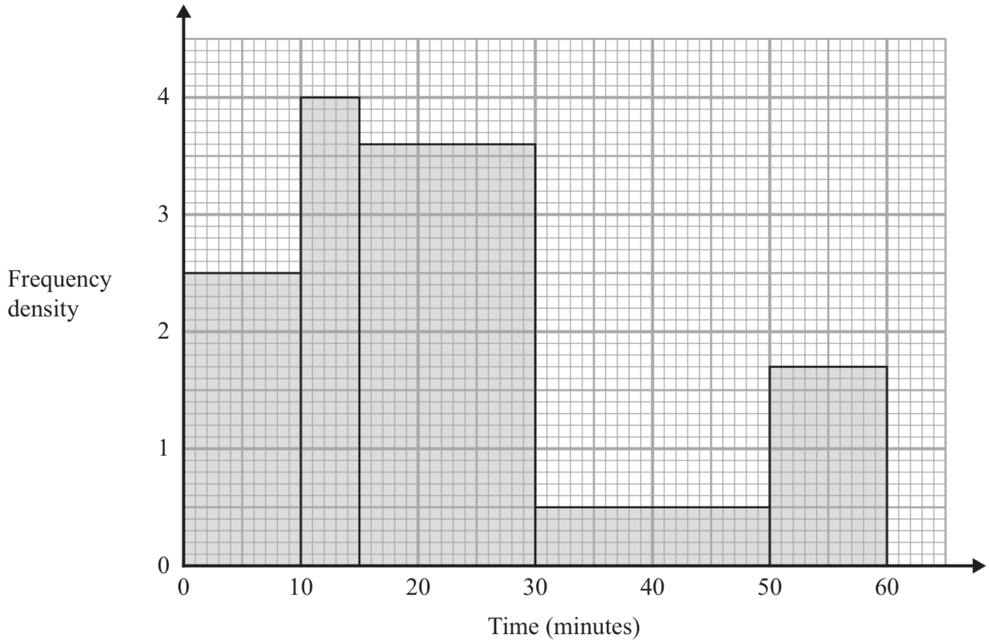
FINAL ANSWER
$f(9) = 6$, and $x < -\frac{9}{2}$ or $x > \frac{9}{2}$.

PAST PAPER QUESTION Q#: 17 Marks: 3

Topic: **Histograms**

Jun 2025 P1H - Jun 2025 | P1H

17 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

....

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Histograms**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Histograms. The tag is correct.

Solution

Total estimated frequency:

$$10(2.5) + 5(4) + 15(3.6) + 20(0.5) + 10(1.7) = 126$$

More than 40 minutes:

$$10(0.5) + 10(1.7) = 22$$

So the proportion is

$$\frac{22}{126} = \frac{11}{63}$$

FINAL ANSWER

$\frac{11}{63}$.

PAST PAPER QUESTION

Q#: 18 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1H - Jun 2025 | P1H

18 y is inversely proportional to x^3

$y = 4$ when $x = 3$

Work out the value of x when $y = 864$

$x =$ ....

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Direct and Inverse Proportion. The tag is correct.

Method

y is inversely proportional to x^3, so

$$y = \frac{k}{x^3}$$

Use $y = 4$ when $x = 3$:

$$4 = \frac{k}{3^3}$$

$$k = 4 \times 27 = 108$$

So

$$y = \frac{108}{x^3}$$

When $y = 864$,

$$864 = \frac{108}{x^3}$$

$$x^3 = \frac{108}{864} = \frac{1}{8}$$

$$x = \frac{1}{2}$$

FINAL ANSWER

$x = \frac{1}{2}$.

PAST PAPER QUESTION Q#: 19 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2025 P1H - Jun 2025 | P1H

19 The equation of a curve is $y = \mathrm{f}(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (8, –12)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = \mathrm{f}(x) + 3$

(.......................... , ...)
(1)

(ii) $y = \mathrm{f}(2x)$

(.......................... , ...)
(1)

(Total for Question 19 is 2 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Correct. This is transforming graph coordinates.

Solution

The minimum point of $y = f(x)$ is

$$(8, 12)$$

For

$$y = f(x) + 5$$

the graph moves up by 5:

$$(8, 12) \to (8, 17)$$

For

$$y = f(2x)$$

the graph is horizontally compressed by scale factor $\frac{1}{2}$, so the x-coordinate is halved:

$$(8, 12) \to (4, 12)$$

FINAL ANSWER

(i) $(8, 17)$ and (ii) $(4, 12)$.

PAST PAPER QUESTION Q#: 20 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

20

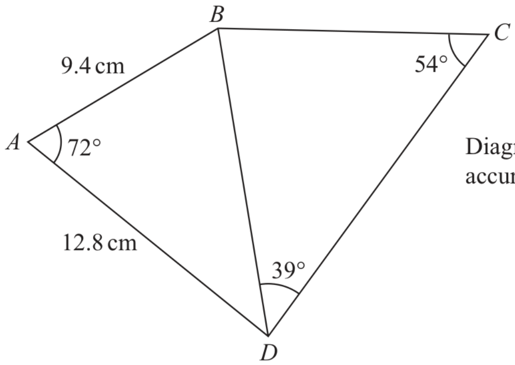

Diagram **NOT** accurately drawn

Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

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Topic check. Moved from Standard & Compound Units to Sine, Cosine Rule & Area of Triangles.

Method

First use triangle ABD.

$$AB = 9.4, \quad AD = 12.8, \quad \angle BAD = 72^\circ$$

Use the cosine rule to find BD:

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

In triangle BCD,

$$\angle BDC = 39^\circ, \quad \angle BCD = 54^\circ$$

Use the sine rule:

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = \frac{13.3356\ldots \sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

Correct to 3 significant figures,

$$BC = 10.4$$

FINAL ANSWER

10.4 cm

PAST PAPER QUESTION **Q#: 20** **Marks: 5**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

20

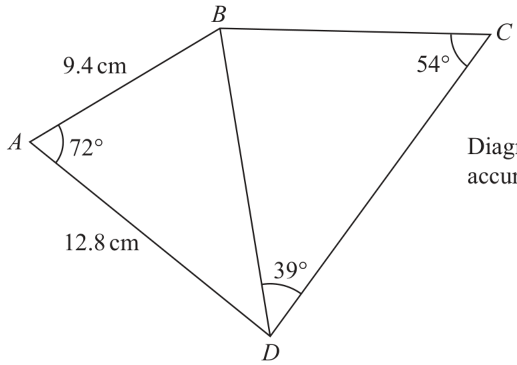

Diagram **NOT** accurately drawn

Work out the length of BC
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Standard & Compound Units to Sine, Cosine Rule & Area of Triangles.

Method

First use triangle ABD.

$$AB = 9.4, \quad AD = 12.8, \quad \angle BAD = 72^\circ$$

Use the cosine rule to find BD:

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

In triangle BCD,

$$\angle BDC = 39^\circ, \quad \angle BCD = 54^\circ$$

Use the sine rule:

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = \frac{13.3356\ldots \sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

Correct to 3 significant figures,

$$BC = 10.4$$

FINAL ANSWER

10.4 cm

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

21 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

There are

$$\binom{20}{3} = 1140$$

ways to choose 3 counters.
Exactly two counters are the same colour:

$$\binom{9}{2}(11) + \binom{7}{2}(13) + \binom{4}{2}(16)$$

$$= 36(11) + 21(13) + 6(16)$$

$$= 396 + 273 + 96 = 765$$

So the probability is

$$\frac{765}{1140} = \frac{51}{76}$$

FINAL ANSWER

$\frac{51}{76}$

PAST PAPER QUESTION

Q#: 21 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

21 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

There are

$$\binom{20}{3} = 1140$$

ways to choose 3 counters.
Exactly two counters are the same colour:

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$$= 36(11) + 21(13) + 6(16)$$

$$= 396 + 273 + 96 = 765$$

So the probability is

$$\frac{765}{1140} = \frac{51}{76}$$

FINAL ANSWER

$\frac{51}{76}$

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

22 Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$
$$y = x + 2$$

Show clear algebraic working.

....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$x^2 + y^2 + 3y = 7$$

$$y = x + 2$$

Substitute:

$$x^2 + (x+2)^2 + 3(x+2) = 7$$

$$x^2 + x^2 + 4x + 4 + 3x + 6 = 7$$

$$2x^2 + 7x + 3 = 0$$

$$(2x+1)(x+3) = 0$$

So

$$x = -\frac{1}{2} \quad \text{or} \quad x = -3$$

Find y:

$$x = -\frac{1}{2} \Rightarrow y = \frac{3}{2}$$

$$x = -3 \Rightarrow y = -1$$

FINAL ANSWER

$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right)$ or $(-3, -1)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

22 Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$
$$y = x + 2$$

Show clear algebraic working.

....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$x^2 + y^2 + 3y = 7$$

$$y = x + 2$$

Substitute:

$$x^2 + (x+2)^2 + 3(x+2) = 7$$

$$x^2 + x^2 + 4x + 4 + 3x + 6 = 7$$

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Find y:

$$x = -\frac{1}{2} \Rightarrow y = \frac{3}{2}$$

$$x = -3 \Rightarrow y = -1$$

FINAL ANSWER

$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right)$ or $(-3, -1)$

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

23

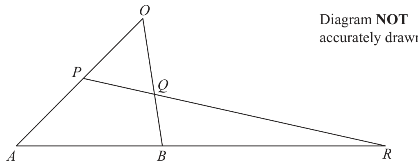

Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of OA
Q is a point on OB

ABR and PQR are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\quad\quad$ $\overrightarrow{OB} = 8\mathbf{b}$

(a) Express $\overrightarrow{AB}$ in terms of **a** and **b**

....
(1)

$AB : BR = 1 : 2$ $\quad\quad$ $\overrightarrow{OQ} = n\mathbf{b}$

(b) Use a vector method to find the value of n

n = ...
(4)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 5

TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Vectors.

Method

For part (a),

$$\vec{OA} = 12\mathbf{a}, \qquad \vec{OB} = 8\mathbf{b}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}$$

For part (b), P is the midpoint of OA, so

$$\vec{OP} = 6\mathbf{a}$$

Given

$$AB : BR = 1 : 2$$

$$\vec{BR} = 2\vec{AB}$$

$$\vec{BR} = 16\mathbf{b} - 24\mathbf{a}$$

Therefore

$$\vec{OR} = \vec{OB} + \vec{BR}$$

$$\vec{OR} = 8\mathbf{b} + 16\mathbf{b} - 24\mathbf{a}$$

$$\vec{OR} = -24\mathbf{a} + 24\mathbf{b}$$

Since P, Q, R are collinear, write Q on PR:

$$\vec{OQ} = \vec{OP} + t(\vec{OR} - \vec{OP})$$

$$\vec{OQ} = 6\mathbf{a} + t(-30\mathbf{a} + 24\mathbf{b})$$

$$\vec{OQ} = (6 - 30t)\mathbf{a} + 24t\mathbf{b}$$

But Q lies on OB, so there is no $\mathbf{a}$ component:

$$6 - 30t = 0$$

$$t = \frac{1}{5}$$

Then

$$\vec{OQ} = 24\left(\frac{1}{5}\right)\mathbf{b}$$

$$\vec{OQ} = \frac{24}{5}\mathbf{b}$$

Since $\vec{OQ} = n\mathbf{b}$,

$$n = \frac{24}{5}$$

FINAL ANSWER
$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}, \quad n = \frac{24}{5}$

PAST PAPER QUESTION Q#: 23 Marks: 5

TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

23

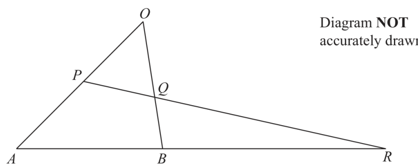

Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of *OA*
Q is a point on *OB*

ABR and *PQR* are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\qquad \overrightarrow{OB} = 8\mathbf{b}$

(a) Express $\overrightarrow{AB}$ in terms of **a** and **b**

....

(1)

$AB : BR = 1 : 2$ $\qquad \overrightarrow{OQ} = n\mathbf{b}$

(b) Use a vector method to find the value of n

$n =$ ....

(4)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Vectors.

Method

For part (a),

$$\vec{OA} = 12\mathbf{a}, \qquad \vec{OB} = 8\mathbf{b}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}$$

For part (b), P is the midpoint of OA, so

$$\vec{OP} = 6\mathbf{a}$$

Given

$$AB : BR = 1 : 2$$

$$\vec{BR} = 2\vec{AB}$$

$$\vec{BR} = 16\mathbf{b} - 24\mathbf{a}$$

Therefore

$$\vec{OR} = \vec{OB} + \vec{BR}$$

$$\vec{OR} = 8\mathbf{b} + 16\mathbf{b} - 24\mathbf{a}$$

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Since P, Q, R are collinear, write Q on PR:

$$\vec{OQ} = \vec{OP} + t(\vec{OR} - \vec{OP})$$

$$\vec{OQ} = 6\mathbf{a} + t(-30\mathbf{a} + 24\mathbf{b})$$

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$$6 - 30t = 0$$

$$t = \frac{1}{5}$$

Then

$$\vec{OQ} = 24\left(\frac{1}{5}\right)\mathbf{b}$$

$$\vec{OQ} = \frac{24}{5}\mathbf{b}$$

Since $\vec{OQ} = n\mathbf{b}$,

$$n = \frac{24}{5}$$

FINAL ANSWER
$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}, \quad n = \frac{24}{5}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Sequences**

Jun 2025 P1H - Jun 2025 | P1H

24 An arithmetic series has 30 terms.

The first term is a
The common difference is d

The 20th term is 123
The sum of the 30 terms is 2880

Work out the value of a and the value of d
Show clear algebraic working.

$a =$ ....

$d =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Sequences**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$u_{20} = a + 19d = 123$$

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$

$$2a + 29d = 192$$

Double the first equation:

$$2a + 38d = 246$$

Subtract:

$$9d = 54$$

$$d = 6$$

$$a + 19(6) = 123$$

$$a = 9$$

FINAL ANSWER

$a = 9,\ d = 6$

PAST PAPER QUESTION Q#: 24 Marks: 5

Topic: **Sequences**

Jun 2025 P1H - Jun 2025 | P1H

24 An arithmetic series has 30 terms.

The first term is a
The common difference is d

The 20th term is 123
The sum of the 30 terms is 2880

Work out the value of a and the value of d
Show clear algebraic working.

$a =$ ....

$d =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Sequences**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$u_{20} = a + 19d = 123$$

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Subtract:

$$9d = 54$$

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$$a = 9$$

FINAL ANSWER

$a = 9, \ d = 6$

PAST PAPER QUESTION

Q#: 25 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

25 *PQRS* is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, –5)

Find an equation of the straight line that passes through the points Q and S
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

....

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$P = (4, 7), \qquad R = (8, -5)$$

The gradient of diagonal PR is

$$\frac{-5 - 7}{8 - 4} = -3$$

The other diagonal QS is perpendicular to PR, so its gradient is

$$\frac{1}{3}$$

The diagonals of a square bisect each other.
The midpoint of PR is

$$\left(\frac{4 + 8}{2}, \frac{7 + (-5)}{2}\right) = (6, 1)$$

So QS passes through $(6, 1)$ with gradient $\frac{1}{3}$.

$$y - 1 = \frac{1}{3}(x - 6)$$

$$y = \frac{1}{3}x - 1$$

Multiply by 3:

$$3y = x - 3$$

FINAL ANSWER

$3y = x - 3$

PAST PAPER QUESTION Q#: 25 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

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Give your answer in the form $ay = bx + c$ where a, b and c are integers.

....

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

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Method

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$$y = \frac{1}{3}x - 1$$

Multiply by 3:

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FINAL ANSWER

$3y = x - 3$